Tobias Mwangi-Hartley

Data Scientist | Machine Learning

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PROFILE

Data scientist with 6 years in analytics and data science, including 4+ years building and operating machine-learning workflows used by product and operations teams. Experienced in Python, feature engineering, forecasting, churn modelling, experimentation and production data workflows using BigQuery, dbt and Vertex AI Workbench. Delivered measurable improvements in forecast accuracy and retention targeting. Developing further production software engineering practices, including automated testing; now moving towards machine learning engineering.

EXPERIENCE

Data Scientist | Thistlebox (meal-kit subscriptions), Edinburgh
Feb 2022–Present

- Built a weekly, six-week-ahead demand forecasting workflow for orders by recipe and region, using LightGBM and calendar, price and menu-position features. Scheduled notebook jobs write forecasts to BigQuery for operations teams to use in Looker.
- Reduced forecast WAPE from 23% to 14%; the resulting workflow was associated with an estimated 9% reduction in food waste.
- Developed an XGBoost churn propensity model, scored weekly in BigQuery, to target retention offers to top-decile customers. A 10% holdout validation showed churn was 2.1 percentage points lower in the treated group.
- Designed and analysed approximately 40 A/B tests per year for growth and product teams; built a sequential testing calculator in Python and Streamlit, used by around 25 colleagues. Wrote the app and Dockerfile; the data platform team handles deployment on Cloud Run.
- Write production SQL in BigQuery and maintain dbt models for the analytics layer.
- Began adding pytest unit tests for forecasting feature code in 2025; coverage is approximately 60%.
- Present results to leadership monthly and delivered internal “Stats for PMs” training three times.

Junior Data Analyst | Firth Energy Retail, Glasgow
Sep 2020–Jan 2022

- Performed customer segmentation and tariff analysis using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

SELECTED PROJECT

Image Classification Course Project | Personal, 2025

- Fine-tuned a small CNN in PyTorch on a public food-images dataset as part of an online deep-learning course; achieved approximately 80% accuracy. Learning project only.
- Technologies: PyTorch, torchvision, Google Colab.

EDUCATION

MSc Statistics, University of the Lothians | 2020 | Distinction
Dissertation: Bayesian hierarchical models for small-area energy demand

BSc Mathematics, University of the Lothians | 2019

TECHNICAL SKILLS

Python (pandas, scikit-learn, LightGBM, XGBoost, statsmodels, pytest), R (tidyverse, brms), SQL (BigQuery), dbt, Git, Jupyter, Looker, Vertex AI Workbench, Streamlit, Docker, Cloud Run

Machine learning and analytics: Feature engineering, time-series forecasting, churn modelling, A/B testing, sequential testing, causal inference, Bayesian statistics, model evaluation